

CBSE Test Paper 04
Chapter 11 Thermal Properties of Matter

1. Heat is a form of **1**
 - a. energy
 - b. solid
 - c. liquid
 - d. gas
 2. The coefficient of area expansion is n times of coefficient of linear expansion. where n equals **1**
 - a. 5
 - b. 2
 - c. 3
 - d. 4
 3. The temperature of a silver bar rises by 10.0°C when it absorbs 1.23 kJ of energy by heat. The mass of the bar is 525 g. Determine the specific heat of silver. **1**
 - a. $0.234 \text{ kJ/kg. }^{\circ}\text{C}$
 - b. $0.254 \text{ kJ/kg. }^{\circ}\text{C}$
 - c. $0.244 \text{ kJ/kg. }^{\circ}\text{C}$
 - d. $0.214 \text{ kJ/kg. }^{\circ}\text{C}$
 4. compared to glass copper expands **1**
 - a. three times less
 - b. five times more
 - c. about the same
 - d. two times less
 5. A 200 cm^3 glass flask is filled to the brim with mercury at 20°C How much mercury overflows when the temperature of the system is raised to 100°C The coefficient of linear expansion of the glass is $0.40 \times 10^{-5}/^{\circ}\text{C}$. Coefficient of volume expansion of mercury is $1.82 \times 10^{-4}/^{\circ}\text{C}$ **1**
 - a. 2.3 cm^3
 - b. 2.5 cm^3
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c. 2.7 cm^3

d. 2.9 cm^3

6. Is the bulb of a thermometer made of diathermic walls or adiabatic walls? **1**
 7. Is it correct to call heat as the energy in transit? **1**
 8. Calorimeters are made of metals, not glass Why? **1**
 9. Briefly explain the concept of thermal equilibrium and temperature. **2**
 10. The triple points of neon and carbon dioxide are 24.57 K and 216.55 K, respectively. Express these temperatures on the Celsius and Fahrenheit scales. **2**
 11. What kind of thermal conductivity and specific heat requirements would you specify for cooking utensils? **2**
 12. The triple points of neon and carbon dioxide are 24.57 K and 216.55 K respectively. Express these temperatures on the Celsius and Fahrenheit scales. **3**
 13. A brass boiler has a base area of 0.15 m^2 and thickness 1.0 cm. It boils water at the rate of 6.0 kg/min when placed on a gas stove. Estimate the temperature of the part of the flame in contact with the boiler. Thermal conductivity of brass = $109 \text{ J s}^{-1}\text{m}^{-1} \text{ K}^{-1}$; Heat of vaporisation of water = $2256 \times 10^3 \text{ J kg}^{-1}$. **3**
 14. A 10 kW drilling machine is used to drill a bore in a small aluminium block of mass 8.0 kg. How much is the rise in temperature of the block in 2.5 minutes, assuming 50% of power is used up in heating the machine itself or lost to the surroundings. Specific heat of aluminium = $0.91 \text{ J g}^{-1} \text{ K}^{-1}$. **3**
 15. A body cools from 80°C to 50°C in 5 minutes. Calculate the time it takes to cool from 60°C to 30°C . The temperature of the surroundings is 20°C . **5**
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